Chapter : Introduction to Vectors

Multiplication of Vectors by Scalars and Addition of Vectors

1. DIRECTED LINE SEGMENTS

Any given portion of a given straight line where for the two end points are distinguished as Initial and Terminal is called a Directed Line Segment. The directed line segment with initial point A and terminal point B denoted by the symbol

$$\overrightarrow{AB}$$

The two end points of a directed line segment are not interchangeable and the directed line segments

$$\overrightarrow{AB} \text{ and } \overrightarrow{BA}$$

Must be directed line segment.

1.1. Length, Support and Sense of a Directed Line Segment

Associated with every directed line segment

$$\overrightarrow{AB},$$

We have its

Length, Support and Sense

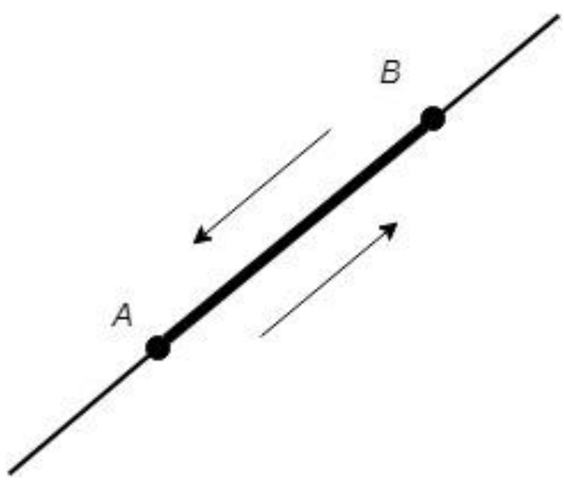


Fig. 1. 1

i. **Length**

The length of $\overrightarrow{AB}$ will be denoted by the symbol

$$\left|\overrightarrow{AB}\right|$$

Clearly we have

$$\left|\overrightarrow{AB}\right| = \left|\overrightarrow{BA}\right|$$

ii. **Support**

The line of unlimited length of which a directed line segment is a part is called its line of support or simply the support.

iii. Sense

The sense of $\overrightarrow{AB}$ is from A to B and that of $\left|\overrightarrow{BA}\right|$ from B to A so that the sense of a directed line segment is from its initial to the terminal point.

The directed line segments

$$\overrightarrow{AB} \text{ and } \overrightarrow{BA}$$

Have the same lengths and supports but different senses (Fig. 1.1).

The question of comparison of the senses of two directed line segments arises only when they have the same or parallel support (Fig.1.2).

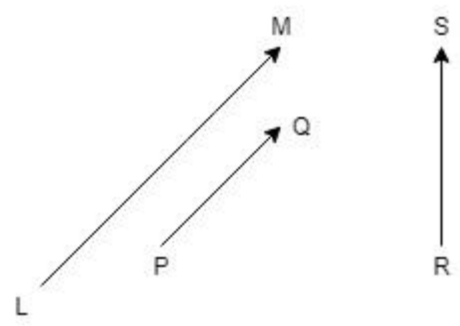


Fig. 1. 2

Two directed line segments having the same or parallel supports may have the same or opposite senses

Ex. How many Directed Line Segments are determined by 2, 3 and 4 given points?

2. VECTORS AND SCALARS

2.1. Vector

Def. *A directed line segment is called vector.*

2.2. **Equality of Two Vectors**

Def. *two vectors are said to be equal if they have*

i. *The same length*
ii. *The same or parallel supports, and*
iii. *The same sense*

It may thus be seen that two different Directed Line Segments may correspond to the same vector.

Thus the vectors

$\overrightarrow{AB}$,$\overrightarrow{CD}$, $\overrightarrow{EF}$ are equal (Fid.1.3).

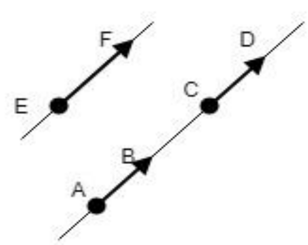


Fig. 1. 3

Two vectors will not be equal if they have different lengths or inclined supports or again, they will not be equal even if they have the same lengths and parallel supports but different senses (Fig. 1.2).

If ABCD is a parallelogram, we have

$$\left|\overrightarrow{AB}\right| = \left|\overrightarrow{DC}\right|$$

and

$$\left|\overrightarrow{BC}\right| = \left|\overrightarrow{AD}\right|$$

Every vector belongs to a class of equal vectors.

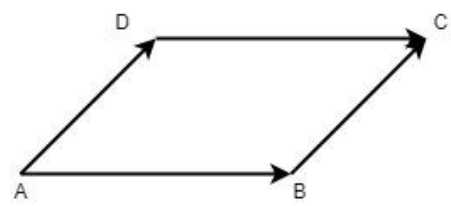


Fig. 1. 4

2.3. Notation for a Vector

A vector is also denoted by a single letter such that **a, b, c** etc.

By using the bold face type so that we may write

$$\boldsymbol{a} = |\overrightarrow{AB}|$$

Then the symbol

$$|\boldsymbol{a}|$$

Denotes the length of the vector, **a**, also called the **Magnitude** of the vector.

2.4. Co-initial Vectors

It is possible to replace a given vector by another equal vector having any given point as its initial point.

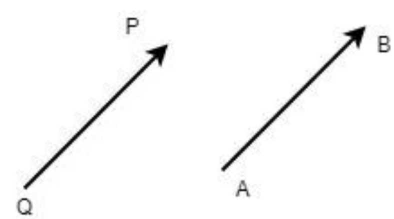


Fig. 1. 5

Thus, if $\overrightarrow{AB}$ be any given vector and *O*, any given point, then by drawing through *O*, a line *OP* parallel to *AB* in the same sense as *AB* of length equal to that of *AB*, we obtain a vector $\overrightarrow{OP}$ equal to $\overrightarrow{AB}$ and with the given point, *O* as the initial point (Fig. 1.5).

2.5. Zero Vector

A vector whose initial and terminal points are coincident is called the **Zero Vector**

The length of the zero vector is zero but it can be thought of as having any line as its line of support.

The zero vector is denoted by the bold face type**.** It will be seen that the zero vector has many properties similar to those of the zero number.

2.6. Scalars

In the following, real numbers will be called scalars. The absolute value of a scalar, *m*, will as usual, be denoted by the symbol $|m|$ so that

$|m| = m$ If ….. and $-m$ if $m < 0$.

3. ALGEBRA OF VECTORS

3.1. Multiplication of Vectors by Scalars

Let, **a,** be any given vector and, *m*, be any given scalar. Then the symbol

$$m\boldsymbol{a},$$

Called the product of the vector, **a**, by the scalar, *m*, is a vector such that

i. The *length* of $m\boldsymbol{a}$, is given by

$$|m\boldsymbol{a}| = |m||\boldsymbol{a}|$$

i.e., the length of the vector, *m* **a** is *m* times or, -*m*, times that of, **a**, according as *m* is positive (including zero) or negative.

Thus, $\boldsymbol{b} = m\boldsymbol{a} \;\; \rightarrow \;\; |\boldsymbol{m}| = |\boldsymbol{b}|/|\boldsymbol{a}|$

ii. The *support* of *m***a,** is the same or parallel, to that of **a.**
iii. The *sense* of *m***a**, is the same or opposite, to that of **a**, according as *m*, is positive or negative.

The symbol, *m***a**, is also sometimes written as**, a***m*, so that the scalar *m*, appears on the right of the vector instead of on the left.

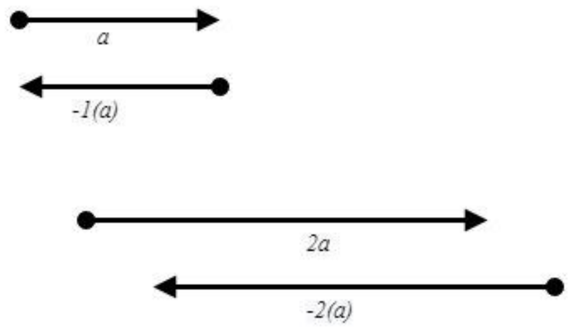


Fig. 1. 6

The following results are immediate consequences of the above definition of the multiplication of vectors by scalars.

I. $(ma)\boldsymbol{a} = m(n\boldsymbol{a})$

where *m*, *n* are any scalars and, **a**, any vector.

II. $0\boldsymbol{a} = \boldsymbol{0}$,

So that the product of a vector, **a**, by the zero scalar is the zero vector.

Here, 0, on the left stands for the zero scalar and **0**, on the right for the zero vector.

III. If two vectors have the same or parallel supports, then each can be thought of as a product of the other by a suitable scalar, the absolute value of the scalar being the ratio of the lengths of the vectors taken in an appropriate order.

Thus, for example, if C be the midpoint of a line AB, we have

$$\overrightarrow{AB} = 2\overrightarrow{AC}, \quad \overrightarrow{BC} = -\frac{1}{2}\overrightarrow{AB}$$

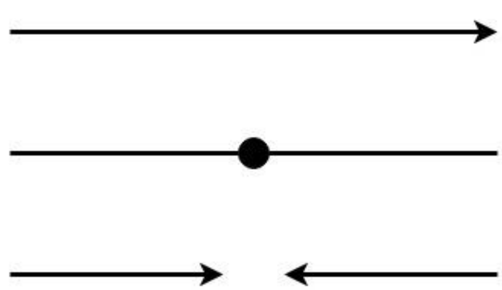


Fig. 1. 7

Two parallel vectors may also be described as collinear.

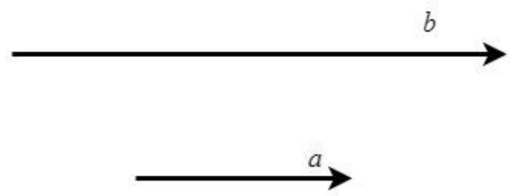


Fig. 1. 8

3.2. Addition Composition

Let **a, b** be two given vectors. Take a point *O*.

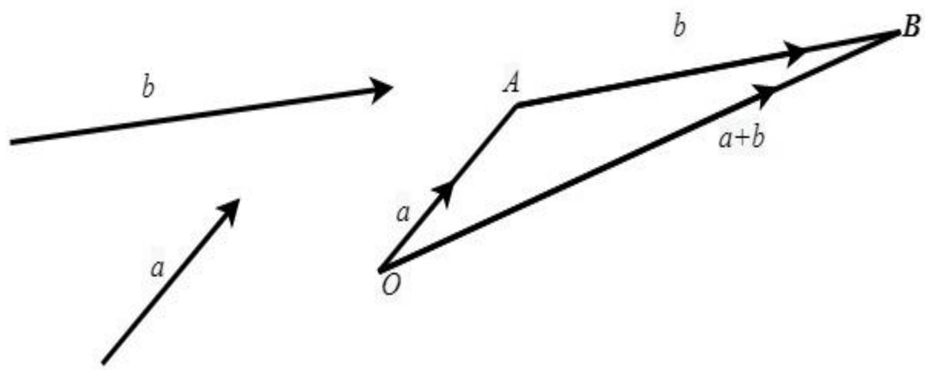


Fig. 1. 9

Let

$$\overrightarrow{OA} = \boldsymbol{a},\ \overrightarrow{AB} = \boldsymbol{b},$$

So that the terminal point of the vector **a** is the initial point of the vector **b**.

The vector

$$\overrightarrow{OB}$$

Is said to be the sum of the vectors, **a** and **b,** and we write

$$\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB} = a + b$$

We have $\boldsymbol{a} + \mathbf{0} = \boldsymbol{a}$**,** for every **a;0,** being the zero vector.

Note. As a matter of logical necessity, we must show that the sum of two vectors is independent of the choice of the point *O*.

Let O, O be any two points and let

$$\overrightarrow{OA} = \boldsymbol{a} = \overrightarrow{O'A'}$$

$$\overrightarrow{AB} = \boldsymbol{b} = \overrightarrow{A'B'}$$

By elementary geometry, we may now deduce that

$$\overrightarrow{O'B'} = \overrightarrow{OB}$$

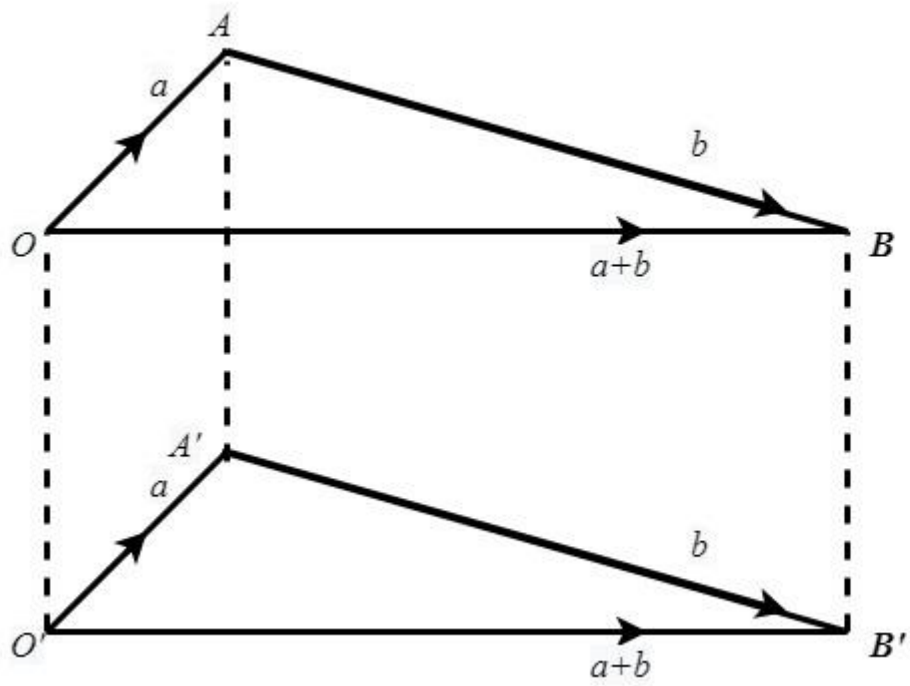


Fig. 1. 10

For this deduction, we have to employ the fact that the lines joining extremities of two equal and parallel straight lines (drawn in the same sense) are themselves equal and parallel.

3.3. Parallelogram Law of Addition of Vectors

Consider a parallelogram OABC. The sum of the vectors $\overrightarrow{OA}$ and $\overrightarrow{OC}$ is the vector $\overrightarrow{OB}$. For $\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB} = \overrightarrow{OA} + \overrightarrow{AC}$.

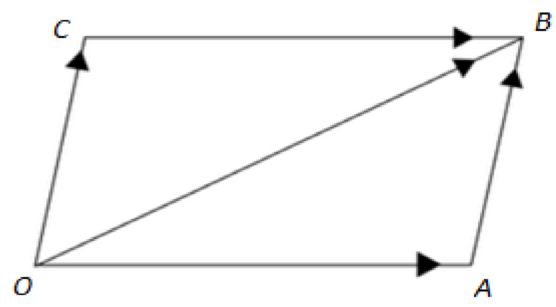


Fig. 1. 11

4. LAWS OF ADDITION COMPOSITION

4.1. **Addition of vector**

Addition of vector is commutative *i.e.*,

$$\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$$

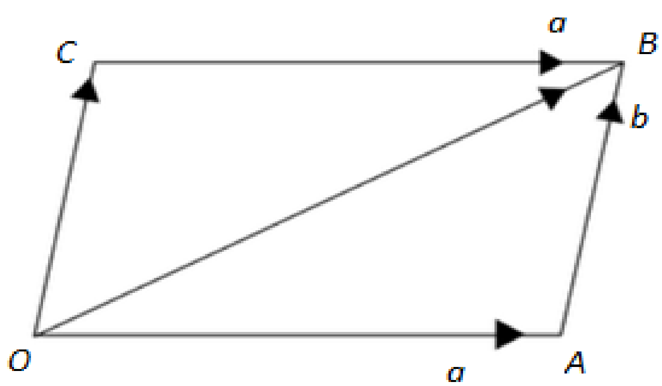


Fig. 1. 12

For any pair of vectors, **a**, **b**.

Let $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{AB} = \mathbf{b}$.

We have $\mathbf{a} + \mathbf{b} = \overrightarrow{OB}$.

Complete the parallelogram OABC having OA and OC as adjacent sides. Then

Multiplication of Vectors by Scalars

$$\overrightarrow{OC} = \overrightarrow{AB} = \mathbf{b},\ \overrightarrow{CB} = \overrightarrow{OA} = \mathbf{a}$$

So that we have

$$\overrightarrow{OB} = \overrightarrow{OC} + \overrightarrow{CB} = \mathbf{b} + \mathbf{a}$$

Hence, **a + b = b + a.**

4.2. Addition of Vectors

Addition of vectors is Associative, i.e.,

$$\mathbf{a} + (\mathbf{b} + \mathbf{c}) = (\mathbf{a} + \mathbf{b}) + \mathbf{c}$$

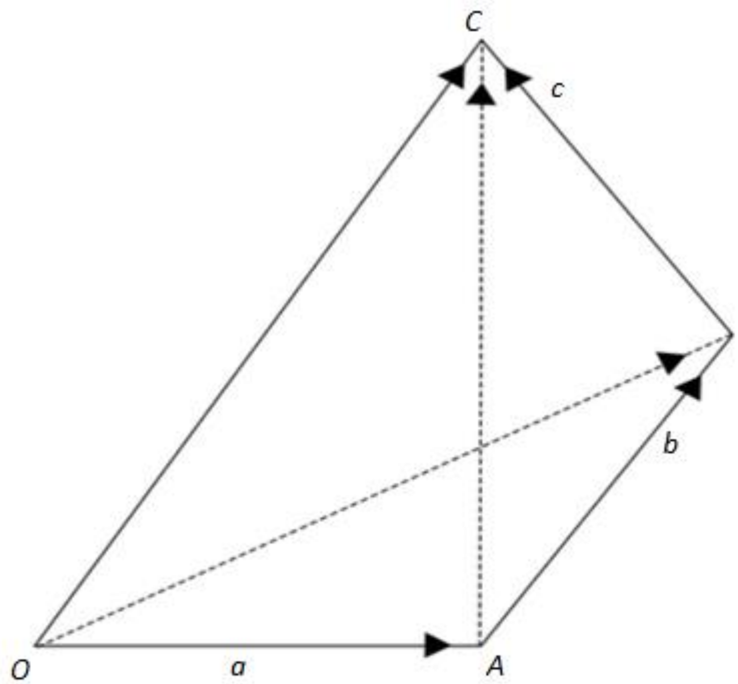


Fig. 1. 13

Where **a, b, c** are any three vectors.

Take any point O. Let

$$\overrightarrow{OA} = \mathbf{a},\ \overrightarrow{AB} = \mathbf{b}, \overrightarrow{BC} = \mathbf{c}.$$

We have

$$\mathbf{b} + \mathbf{c} = \overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$$

$$\Rightarrow \quad \mathbf{a} + (\mathbf{b} + \mathbf{c}) = \overrightarrow{OA} + \overrightarrow{AC} = \overrightarrow{OC}\,.$$

Again

$$\mathbf{a} + \mathbf{b} = \overrightarrow{OA} + \overrightarrow{AB} = \overrightarrow{OB}$$

$$\Rightarrow \quad (\mathbf{a} + \mathbf{b}) + \boldsymbol{c} = \overrightarrow{OB} + \overrightarrow{AC} = \overrightarrow{OC}\,.$$

Thus

$$(\mathbf{a} + \mathbf{b}) + \boldsymbol{c} = \overrightarrow{OC} = \mathbf{a} + (\mathbf{b} + \mathbf{c}).$$

In view of the equality of the vectors.

$$(\mathbf{a} + \mathbf{b}) + \mathbf{c}\,,\ \mathbf{a} + (\mathbf{b} + \mathbf{c})$$

We denote each of these equal vectors by

$$\mathbf{a} + \mathbf{b} + \mathbf{c}.$$

4.3. Negative of a vector

For every vector **a,**

$$\mathbf{a} + (-1)\,\mathbf{a} = \mathbf{0}.$$

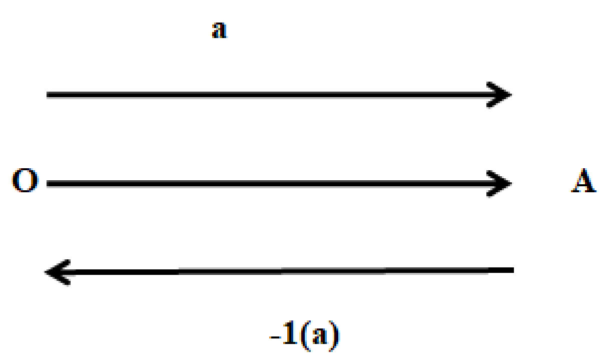


Fig. 1. 14

If $\overrightarrow{OA} = \mathbf{a}$, we have, according to the definition of the multiplication of vectors by scalars,

$$\overrightarrow{AO} = (-1)\mathbf{a}.$$

Thus,

$$\mathbf{a} + (-1)\,\mathbf{a} = \overrightarrow{OA} + \overrightarrow{AO} = \overrightarrow{OO} = \mathbf{0}.$$

On account of this property. The vector (-1) **a** is called the negative of the vector **a,** and we write

$$\mathbf{-a = (-1)\ a}$$

So that the relation **a** + (-1) **a = 0,** may also be re-written as

$$\mathbf{a + (-a) = 0.}$$

5. RELATIONS BETWEEN THE TWO COMPOSITIONS

We shall now consider the relations between the two line compositions considered in 3.1 and 3.2.

(i) m (**a** + **b**) = m**a** + m**b.**
(ii) (m + n) **a** = m**a** + n**a.**

Where m , n are scalars and **a, b** are vectors

(i) Let m , be positive and let

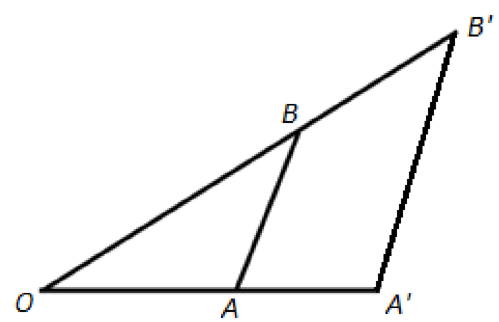


Fig. 1. 15

$$\overrightarrow{OA} = \boldsymbol{a}\,, \overrightarrow{AB} = \boldsymbol{b},$$

We have

$$\overrightarrow{OB} = \mathbf{a} + \mathbf{b}.$$

Let $\overrightarrow{OA'}$= m**a.**

Through A' draw line parallel to AB and in the same sense as to meet OB in B'. With the help of Elementary Geometry, we have

$$A'B' = m\,\overrightarrow{AB} \text{ and } OB' = m\,OB\,.$$

Thus, we have

$$A'B' = m\,\overrightarrow{AB} = m\,\mathbf{a} + m\,\mathbf{b}$$

And

$$OB' = m\,OB = m\,(\,\mathbf{a} + \mathbf{b}\,).$$

It follows that

$$m\,(\,\mathbf{a} + \mathbf{b}\,) = \;OB' = \;OA' + A'B' = \;m\,\mathbf{a} + m\,\mathbf{b}.$$

The result may similarly be proved with the help of the accompany figure 16, when m is negative.

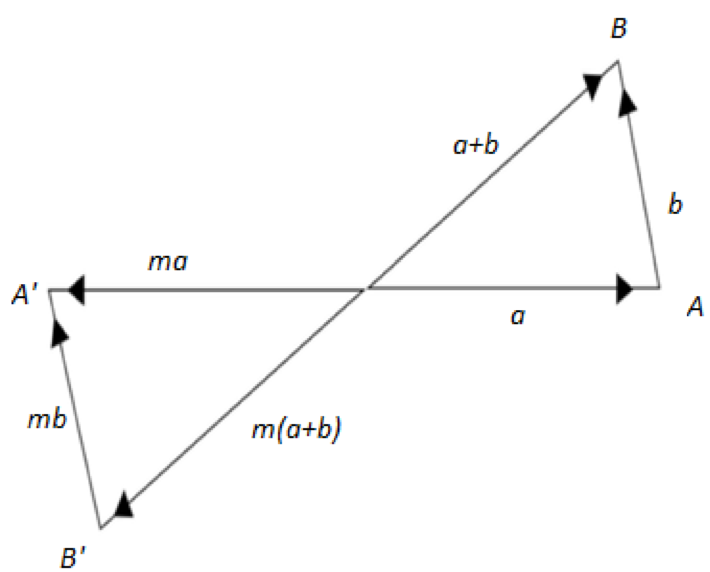


Fig. 1. 16

(ii) The proof is a simple development of the definitions.

6. SUBTRACTION

6.1. Difference of Two Vector.

Definition

If **a** , **b** be any two given vectors, then we write

$$\mathbf{a} + (-\mathbf{b}) = \mathbf{a} - \mathbf{b}$$

and call the composition Subtraction.

Thus , we have , in particular

$$\mathbf{a} - \mathbf{a} = \mathbf{a} + (-\mathbf{a}) = \mathbf{a} + (-1)\,\mathbf{a} = \mathbf{0}.$$

It is important to see that

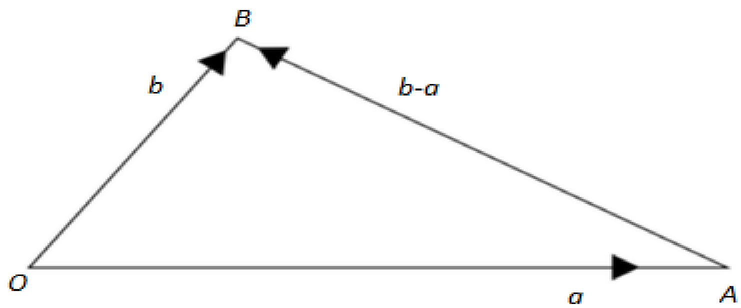


Fig. 1. 17

$\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b} \Longrightarrow \overrightarrow{AB} = \mathbf{b} - \mathbf{a}.$

In fact We have

$$\mathbf{b} - \mathbf{a} = \mathbf{b} + (-\mathbf{a})$$

$$= \overrightarrow{OB} + \overrightarrow{AO} = \overrightarrow{AO} + \overrightarrow{OB} = \overrightarrow{AB}.$$

1.6.1 . $-(\mathbf{a} + \mathbf{b}) = -\mathbf{a} - \mathbf{b}.$ We have

$$[\,(-\mathbf{a}) + (-\mathbf{b})\,] + [\,\mathbf{b} + \mathbf{a}\,] = (-\mathbf{a}) + [\,(-\mathbf{b}) + \mathbf{b}\,] + \mathbf{a}$$

$$= (-\mathbf{a}) + \mathbf{0} + \mathbf{a}$$

$$= (-\mathbf{a} + \mathbf{a}\,) + \mathbf{0} = \mathbf{0}.$$

Thus , $$(-\mathbf{a}) + (-\mathbf{b}) = -(\mathbf{a} + \mathbf{b})$$

$$\Longrightarrow (\,-\mathbf{a} + \mathbf{b}\,) = -\mathbf{a} - \mathbf{b}.$$

The proof of the result may be also directly obtained with the help of fig.1.18 given below

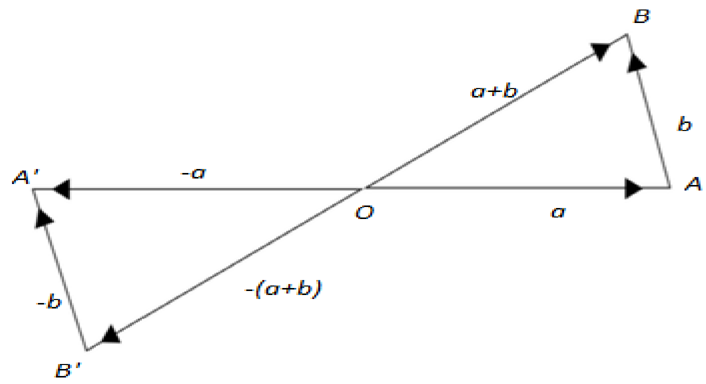


Fig. 1. 18

7. A VECTOR EQUATION

If **a , b** be two given vectors, then the vector equation

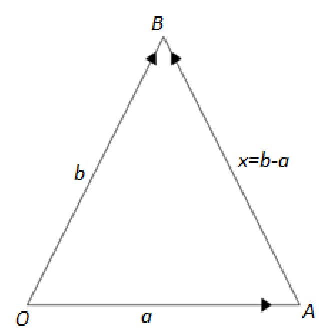


Fig. 1. 19

$$\mathbf{a} + \mathrm{x} = \mathbf{b}.$$

is satisfied by one and only vector. Viz,

$$\mathrm{x} = \mathbf{b} - \mathbf{a}.$$

we have

$\Leftrightarrow \quad \mathbf{a} + \mathrm{x} = \mathbf{b}$

$\Leftrightarrow (-\mathbf{a}) + (\mathbf{a} + \mathrm{x}) = (-\mathbf{a}) + \mathbf{b}$

$\Leftrightarrow [(-\mathbf{a}) + \mathbf{a}] + \mathrm{x} = \mathbf{b} + (-\mathbf{a})$

$\Leftrightarrow \quad \mathbf{0} + \mathrm{x} = \mathbf{b} - \mathbf{a}$

$\Leftrightarrow \quad \mathrm{x} = \mathbf{b} - \mathbf{a}$

8. A SUMARY OF THE BASIC PROPERTIES OF THE LINE COMPOSITIONS

We catalogue below the basic laws of the addition of vectors and multiplication of vectors by scalars, *i.e* , of the linear compositions.

Set of all the vectors will be denoted by **V** and the set of all real numbers by **R.**

I. $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a} \ \forall \ \mathbf{a}, \mathbf{b} \in \mathbf{V};$

II. $\mathbf{a} + (\mathbf{b} + \mathbf{c}) = (\mathbf{a} + \mathbf{b}) + \mathbf{c} \ \forall \ \mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbf{V};$

III. $\exists$ a vector, viz, **0** such that

$$\mathbf{a} + \mathbf{0} = \mathbf{a} = \mathbf{0} + \mathbf{a} \ \forall \ \mathbf{a} \in \mathbf{V};$$

IV. To each $\mathbf{a} \in \mathbf{V}$ there corresponds $\mathbf{b} \in \mathbf{V}$ such that

$$\mathbf{a} + \mathbf{b} = \mathbf{0} = \mathbf{b} + \mathbf{a};$$

V. $(m\,n)(\mathbf{a}) = m(n\,\mathbf{a}) \ \forall \ \mathbf{a} \in \mathbf{V}$ and $\forall \ m, n \in \mathbf{R};$

VI. $I(\mathbf{a}) = \mathbf{a} \ \forall \ \mathbf{a} \in \mathbf{V};$

VII. $m(\mathbf{a} + \mathbf{b}) = m\mathbf{a} + m\mathbf{b} \ \forall \ m \in \mathbf{R}$ and $\forall \ \mathbf{a}, \mathbf{b} \in \mathbf{V};$

VIII. $(m + n)\,\mathbf{a} = m\mathbf{a} + n\,\mathbf{a}. \ \forall \ m \in \mathbf{R}$ and $\forall \ \mathbf{a}, \mathbf{b} \in \mathbf{V};$

9. LINEAR COMBINATIONS

A vector **r,** is said to be a linear combination of the vectors **a , b , c…………** etc. if there exist scalars x, y, z etc. such that

$$\mathbf{r} = x\mathbf{a} + y\mathbf{b} + z\mathbf{c}$$

Thus , for example, the vectors

$$2\mathbf{a} + \mathbf{b} - 4\mathbf{c}, \quad \mathbf{a} + 2\mathbf{b} - 3\mathbf{c}$$

Are linear combination of the vectors **a , b , c.**

A linear combination of vctors involves the two linear composition of the addition of vectors and the multiplication of vectors by scalar in the following linear combinations.

$$x\mathbf{a}, \quad x\mathbf{a}+y\mathbf{b}, \quad x\mathbf{a}+y\mathbf{b}+z\mathbf{c}$$

will be of special interest to us.

9.1. Parallel Vectors. Collinear Vectors

Of the two vectors having the same or parallel supports, each is a linear combination of the order. Thus, if **a , b** be two parallel vectors, then there exists a scalar x such that

$\mathbf{b} = x\mathbf{a}$ or $\mathbf{a} = (1/x)\mathbf{b}.$

Parallel vectors are also called collinear vectors, for the support of parallel coinitial vectors is the same line.

9.2. Definition Coplanar Vectors

A set of vectors is said to be coplanar, if their supports are parallel to the same plane, i.e., if there exists a plane parallel to the supports of each of the vectors.

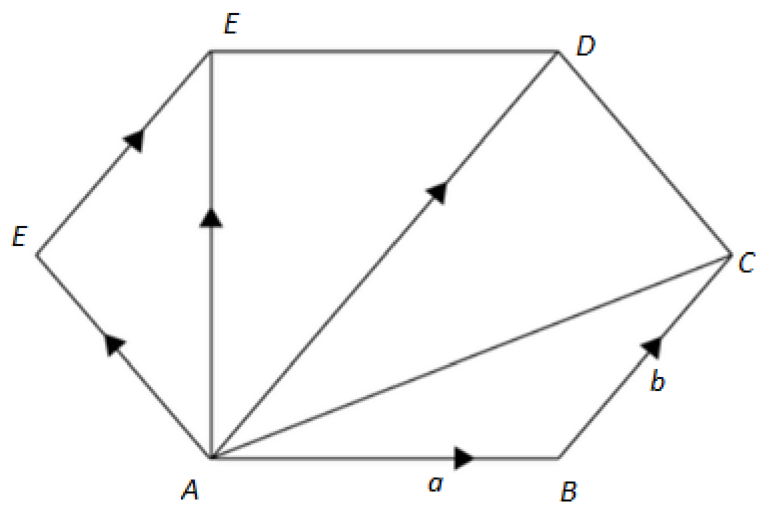
The supports of coplanar co-initial vectors are coplanar.

The vector x**a**+y**b** which is a linear combination of the vectors **a , b** is coplanar with **a** and **b.**

EXAMPLE

ABCDEF is a regular hexagon. Let $\overrightarrow{AB} = a$ and $\overrightarrow{BC} = b$. Find the vector determined by the other four sides taken in order. Also express the vectors $\overrightarrow{AC}, \overrightarrow{AD}, \overrightarrow{AF}, \overrightarrow{AE}, \overrightarrow{CE}$ in terms of **a** and **b**

Solution



$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC} = \mathbf{a} + \mathbf{b}$

∴ AD is parallel and double of BC.

∴ $\overrightarrow{AD}$ = 2**b**

In $\triangle$ ACD,

$$\overrightarrow{AC} + \overrightarrow{CD} = \overrightarrow{CD}$$

$$\Longrightarrow \overrightarrow{CD} = \overrightarrow{AD} - \overrightarrow{AC}$$

$$=2\mathbf{b} - (\mathbf{a} + \mathbf{b})$$

$$= \mathbf{b} - \mathbf{a}.$$

$$\overrightarrow{AF} = \overrightarrow{CD} = \mathbf{b} - \mathbf{a}$$

Now,

$$\overrightarrow{DE} = \overrightarrow{BA} = -\mathbf{a}$$

$$\overrightarrow{EF} = \overrightarrow{CB} = -\mathbf{b}$$

$$\overrightarrow{FA} = \overrightarrow{DC} = -(\mathbf{b} - \mathbf{a}) = \mathbf{a} - \mathbf{b}$$

Again,

$$\overrightarrow{AE} = \overrightarrow{AD} + \overrightarrow{DE} = 2\mathbf{b} + (-\mathbf{a}) = 2\mathbf{b} - \mathbf{a}$$

And $\overrightarrow{CE} = \overrightarrow{CD} + \overrightarrow{DE} = \mathbf{b} - \mathbf{a} + (-\mathbf{a})$

$$= \mathbf{b} - 2\mathbf{a}$$

Scalar Product

Introduction

The concept of the *Scalar Product* of two vectors, as a result of which a scalar is associated to any given pair of vectors, will be introduced in this chapter. The justification for the use of the word *Product* lies in the fact that the so-called scalar product of two vectors is a scalar proportional to the length of each of the two factor vectors and also obeys the Distributive Law like the product of numbers. Also one very important use of the notion of scalar product is that it enables the lengths of vectors and the angles between pairs of vectors to be expressed in terms of the same and thus provides an analytical tool for the study of *Metric Geometry*.

It will also be possible to obtain various formulae of the Three Dimensional Cartesian Geometry simply as a result of translation in terms of Cartesian co-ordinates of the corresponding Vector notation formulae.

Some well-known properties of tetrahedral amenable to treatment by scalar products will also be obtained.

SCALAR PRODUCT

Scalar Product of Two Vectors

Definition

The scalar product of two vectors **a, b** *is the scalar*

$$|a||b|\cos\theta$$

Where, θ, is the angle, between the vectors **a** and **b.**

Also the scalar product of any vector with the zero vector is, by definition, the scalar zero.
The scalar product of the vectors **a, b** is denoted by the symbol

$$\boldsymbol{a}.\boldsymbol{b}$$

and is on this account, also called the *Dot product* of the two vectors **a, b**. Thus, we have, by definition,

$$\boldsymbol{a}.\boldsymbol{b} = |\boldsymbol{a}||\boldsymbol{b}|\cos\theta$$

θ being the angle between the vectors **a, b.**

Note

It may be easily seen that the scalar product of two vectors remains unaltered when they are replaced by vectors equal to the same so that

$$\boldsymbol{a} = \boldsymbol{a}' \text{and } \boldsymbol{b} = \boldsymbol{b}' \quad \Rightarrow \quad \boldsymbol{a}.\boldsymbol{b} = \boldsymbol{a}'.\boldsymbol{b}'$$

Ex. D is the mid-point of the side BC of a triangle ABC; Show that

$$\overrightarrow{DA}.\overrightarrow{DB}+\overrightarrow{DA}.\overrightarrow{DC}=0$$

Sign of the Scalar Product

Definition

*If **a, b** be two non-zero vectors, then the scalar product*

$$\boldsymbol{a}.\boldsymbol{b}=|\boldsymbol{a}||\boldsymbol{b}|\cos\theta$$

is positive, negative or zero, according as the angle, θ , between the vectors is acute, obtuse or right.

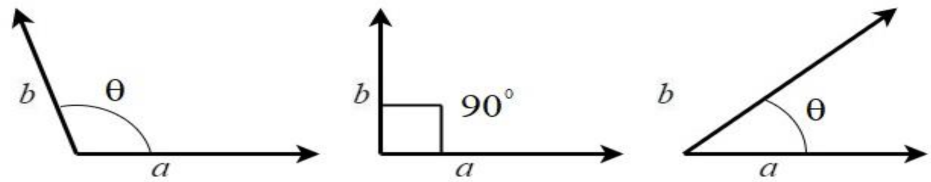


Fig.1. 1

Thus

$$\theta\ is\ acute \implies \cos\theta>0 \implies \boldsymbol{a}.\boldsymbol{b}>0$$

$$\theta\ is\ right \implies \cos\theta=0 \implies \boldsymbol{a}.\boldsymbol{b}=0$$

$$\theta\ is\ obtuse \implies \cos\theta<0 \implies \boldsymbol{a}.\boldsymbol{b}<0$$

We notice that if **a, b** be two vectors, then their scalar product will be zero if and only if, either one at least of the two vectors is the zero vector or the two vectors are at right angles to each other. Thus

$$\boldsymbol{a}.\boldsymbol{b}=0 \implies \boldsymbol{a}=\boldsymbol{0}\ \ \boldsymbol{or}\ \ \boldsymbol{b}=\boldsymbol{0}\ \ \boldsymbol{or}\ \ \boldsymbol{\theta}=\boldsymbol{90^0}$$

θ being the angle between the vectors **a, b.**

It is very important to remember that the *scalar product of two non-zero vectors is zero if and only if they are at right angles to each other.*

Length of a Vector as a Scalar Product

If **a** be any vector, then the scalar product **a.a** of **a** with itself is given by

$$\boldsymbol{a}.\boldsymbol{a}=|\boldsymbol{a}||\boldsymbol{a}|\cos 0=\ |a|^2$$

Thus, the length $|\boldsymbol{a}|$ *of any vector,* $\boldsymbol{a}$*, is the non-negative square root* $\sqrt{(\boldsymbol{a}.\boldsymbol{a})}$*of the scalar product **a.a***

A convention. a.a will be denoted as $\boldsymbol{a^2}$**,** so that $\boldsymbol{a^2}$**,** is a scalar which equals the square of the length of **a.**

Angle between Two Vectors in terms of Scalar Products

If θ be the angle between two non-zero vectors, a, b, we have

$$\boldsymbol{a}.\boldsymbol{b} = |\boldsymbol{a}||\boldsymbol{b}|\cos\theta$$

$$\Longrightarrow \qquad \theta = \cos^{-1}\frac{\boldsymbol{a}.\boldsymbol{b}}{|\boldsymbol{a}||\boldsymbol{b}|} = \cos^{-1}\frac{\boldsymbol{a}.\boldsymbol{b}}{(+\sqrt{\boldsymbol{a}.\boldsymbol{a}})(+\boldsymbol{b}.\boldsymbol{b})}$$

EXERCISES

1. Is it true that
   i. $\boldsymbol{a}.\boldsymbol{b} = \boldsymbol{a}.\boldsymbol{c} \quad \Longrightarrow \quad b = c$?
   ii. $\boldsymbol{a}.\boldsymbol{b} = \boldsymbol{0} \quad \Longrightarrow \quad a \perp b$?

2. Give two points *O*, A; identity the locus of the point P in each of the following cases:
   i. $\overrightarrow{OP}.\overrightarrow{OA} > 0$
   ii. $\overrightarrow{OP}.\overrightarrow{OA} = 0$
   iii. $\overrightarrow{OP}.\overrightarrow{OA} < 0$

3. Given two points A, B, what is the locus of the point P such that
   i. $\overrightarrow{PA}.\overrightarrow{PB} < 0$
   ii. $\overrightarrow{PA}.\overrightarrow{PB} = 0$
   iii. $\overrightarrow{PA}.\overrightarrow{PB} > 0$

4. Given that **a, b** are two vectors of lengths 1 and 2 respectively and $\boldsymbol{a}.\boldsymbol{b} = -\sqrt{\boldsymbol{3}}$**.** What is the angle between the vectors **a** and **b ?**

PROPERTIES OF SCALAR PRODUCT

We shall now obtain some basic properties of scalar product

Commutativity

$$\boldsymbol{a}.\boldsymbol{b} = \boldsymbol{b}.\boldsymbol{a}$$

For every pair of vectors **a, b.**

This property is obvious from the definition

$$\boldsymbol{a}.(-\boldsymbol{b}) = -(\boldsymbol{a}.\boldsymbol{b}); \quad (-\boldsymbol{a}).(-\boldsymbol{b}) = \boldsymbol{a}.\boldsymbol{b}$$

for every pair of vectors **a**, **b**.
The proof is simple.[Refer Fig.1.2]

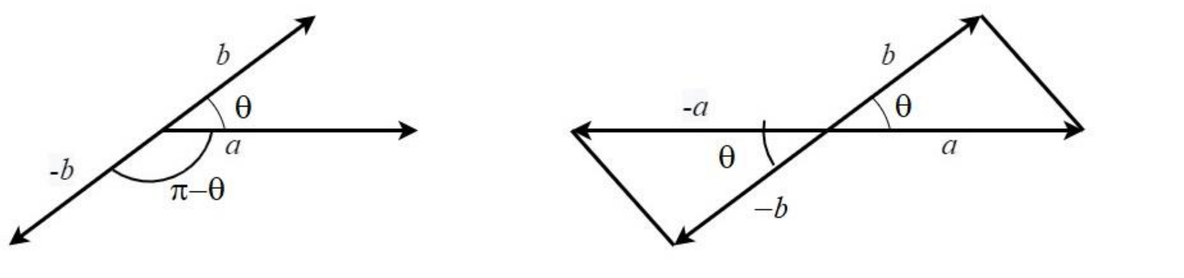


Fig.1. 2

ma.nb* $=$ *mn* (a, b) *where a. b are any vectors and m, n are any scalars

Let *m*, *n* be both positive, so that the angle between *m**a*** and *n**b*** is the same as that between **a** and **b**

We have

$$m\boldsymbol{a}.n\boldsymbol{b} = |m\boldsymbol{a}||n\boldsymbol{b}|\cos\theta$$

$$= |m||\boldsymbol{a}||n||\boldsymbol{b}|\cos\theta$$

$$= mn|\boldsymbol{a}||\boldsymbol{b}|\cos\theta$$

$$= mn(\boldsymbol{a}.\boldsymbol{b})$$

Let now *m* be positive and *n* be negative so that the angle between *m**a*** and *n**b*** is π–θ; θ being the angle between **a** and **b.**

We have

$$m\boldsymbol{a}.n\boldsymbol{b} = |m\boldsymbol{a}||n\boldsymbol{b}|\cos(\pi - \theta)$$

$$= |m||\boldsymbol{a}||n||\boldsymbol{b}|(-\cos\theta)$$

$$= -mn|\boldsymbol{a}||\boldsymbol{b}|\,(-\cos\theta)$$

$$= mn|\boldsymbol{a}||\boldsymbol{b}|\cos\theta$$

$$= mn(\boldsymbol{a}.\boldsymbol{b})$$

Other cases may be similarly disposed of.

If the Scalar Product of a vector, r, with each of three non-coplanar vectors is zero, then, r, must be the zero vector

This follows from the fact that no non-zero vector can be perpendicular to each of three non-coplanar vectors.

In particular, if a vector, **r**, is perpendicular to every vector, then **r** must be the zero vector.

Let b $= \overrightarrow{PQ}$ and let L, M be the feet of the perpendiculars for P and Q on the support AB of the vector a. The projection of b upon AB $= LM = |b|\cos\theta$

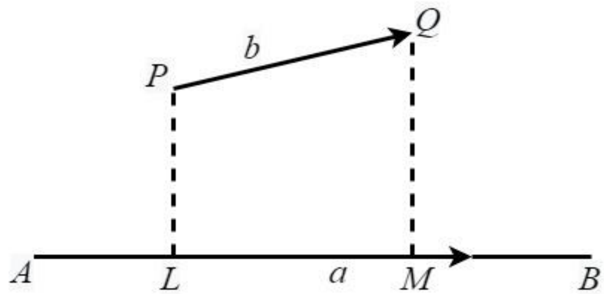


Fig.1. 3

*Thus the scalar product **a. b** is the product of the length of **a**, with the projection of **b**, upon **a**, taken with the proper sign.*

Distributivity

Scalar multiplication of vectors distributes the addition of vectors. We have to prove that

$$\boldsymbol{a}.(\boldsymbol{b}+\boldsymbol{c}) = \boldsymbol{a}.\boldsymbol{b}+\boldsymbol{a}.\boldsymbol{c}$$

for all vectors **a,b,c.**

This property is an immediate consequence of the fact that the *projection of the sum of two vectors on any line is equal to the sum of their projections on the same line*.

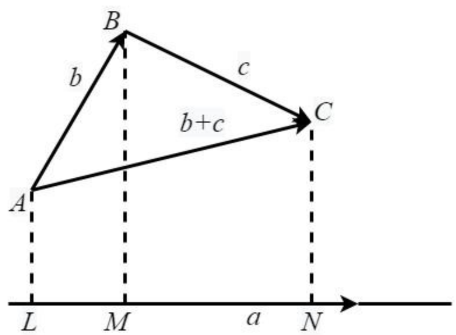


Fig.1. 4

Let

$$\overrightarrow{AB} = b,\ \ \overrightarrow{BC} = c \ so \ that\ \overrightarrow{AC} = b + c$$

Let *L, M, N* be the feet of the perpendiculars from *A, B, C* on the line of support of the vector **a**
Let $pr\boldsymbol{b},\ pr\boldsymbol{c},\ pr(\boldsymbol{b}+\boldsymbol{c})$ denote the projections of **b, c** and **b+c** on **a.**
We have

$$pr\boldsymbol{b} = LM,\ pr\boldsymbol{c} = MN,\ pr(\boldsymbol{b}+\boldsymbol{c}) = LN,$$

So that $$pr(\boldsymbol{b}+\boldsymbol{c}) = pr\boldsymbol{b} + pr\boldsymbol{c}$$

Thus $$\boldsymbol{a}.(\boldsymbol{b}+\boldsymbol{c}) = |\boldsymbol{a}|pr(\boldsymbol{b}+\boldsymbol{c})$$

$$= |\boldsymbol{a}|(pr\boldsymbol{b} + pr\boldsymbol{c})$$

$$= |\boldsymbol{a}|pr\boldsymbol{b} + |\boldsymbol{a}|pr\boldsymbol{c}$$

$$= \boldsymbol{a}.\boldsymbol{b} + \boldsymbol{a}.\boldsymbol{c}$$

It may also be shown that

$$\boldsymbol{a}.(\boldsymbol{b} - \boldsymbol{c}) = \boldsymbol{a}.\boldsymbol{b} - \boldsymbol{a}.\boldsymbol{c}$$

We have

$$\boldsymbol{a}.(\boldsymbol{b} - \boldsymbol{c}) = \boldsymbol{a}.[\boldsymbol{b} + (-\boldsymbol{c})]$$

$$= \boldsymbol{a}.\boldsymbol{b} + \boldsymbol{a}.(-\boldsymbol{c})$$

$$= \boldsymbol{a}.\boldsymbol{b} + [-(\boldsymbol{a}.\boldsymbol{c})]$$

$$= \boldsymbol{a}.\boldsymbol{b} - \boldsymbol{a}.\boldsymbol{c}$$

Some Simple Identities

i. $(a + b).(a - b) = a.a - b.b = a^2 - b^2$

ii. $(a + b)^2 = (a + b).(a + b)$
$= a.a + 2a.b + b.b$
$= a^2 + 2a.b + b^2$

iii. $(a - b)^2 = (a - b).(a - b)$
$= a.a - 2a.b + b.b$
$= a^2 - 2a.b + b^2$

These identities are simple consequences of the fact that the scalar multiplication distributes the addition of vectors and of the other results in 19

We may also note that

$$\boldsymbol{a}.\boldsymbol{b} = \frac{1}{4}(|\boldsymbol{a} + \boldsymbol{b}|^2 - |\boldsymbol{a} - \boldsymbol{b}|^2)$$

So that the scalar product of two vectors is expressed in terms of the lengths of their sum and difference

APPLICATIONS

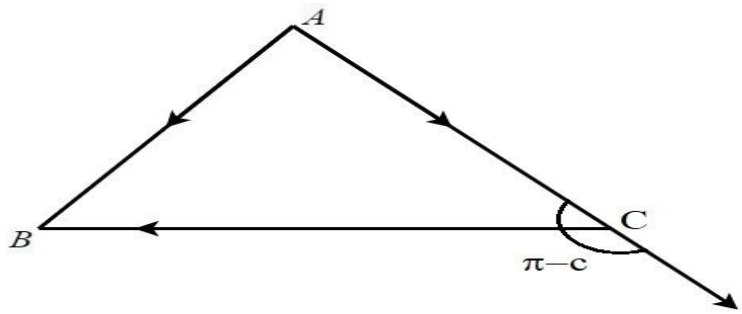
Cosine Formula for Triangles

To prove that for any triangle ABC

$$c^2 = b^2 + a^2 - 2ab\cos C$$

In the usual notation of plane Trigonometry

Solution



We have

$$\overrightarrow{AC} + \overrightarrow{CB} = \overrightarrow{AB}$$

$$\Rightarrow \qquad (\overrightarrow{AC} + \overrightarrow{CB}).(\overrightarrow{AC} + \overrightarrow{CB}) = \overrightarrow{AB}.\overrightarrow{AB}$$

$$\Rightarrow \qquad AC^2 + CB^2 + 2\overrightarrow{AC}.\overrightarrow{CB} = AB^2$$

$$\Rightarrow \qquad b^2 + a^2 - 2ab\cos C = c^2$$

Projection Formula for Triangles

To prove that for any triangle ABC

$$c = b\cos A + a\cos B$$

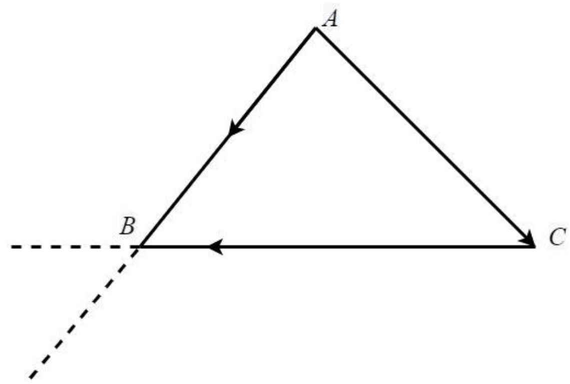


Fig.1. 5

Solution

We have $\qquad \overrightarrow{AC} + \overrightarrow{CB} = \overrightarrow{AB}$

$$\Rightarrow \qquad \overrightarrow{AC}.\overrightarrow{AB} + \overrightarrow{CB}.\overrightarrow{AB} = \overrightarrow{AB}.\overrightarrow{AB}$$

$$\Rightarrow \qquad bc\cos A + ca\cos B = c^2$$

$$\Rightarrow \qquad b\cos A + a\cos B = c$$

Vector Product and Scalar Triple Product

Introduction

The concept of the *Vector product* of two vectors, as a result of which a well-defined vector is associated to a given ordered pair of vectors, will be introduced in this chapter. Like Scalar products, Vector products are also proportional to the lengths of the factor vectors and obey Distributive Laws. It should be, however, remembered that quite a number of the properties of Scalar and Vector products are different from those of ordinary products of numbers.

The notion of vector products greatly facilitates the study of a type of metrical relations such as those involving areas and volumes. Also in those contexts where we require dealing with vectors perpendicular to two given vectors, vector products play an important part.

In addition to Scalar and Vector products, *Mixed product*

$$\boldsymbol{a} \times \boldsymbol{b}.\boldsymbol{c},$$

Known as Scalar triple product, also plays an important part. We shall also consider this in this chapter

The volume of a parallelepiped is given as a scalar triple product and the vanishing of a scalar triple product provides a very neat form of condition for three vectors to be coplanar and four points to be coplanar. The part of the Algebra of vectors developed in this chapter will also provide important tools for the solution of vector Equations.

RIGHT-HANDED AND LEFT-HANDED VECTOR TRIADS

A screw experiences a motion of translation when it is rotated and, as such, we can distinguish a screw as being right-handed or left-handed by the direction of its translation when it is rotated in any given manner. Also we shall now associate a screw with any given ordered vector triad and thus be enabled to distinguish any given vector triad also as being right-handed or left-handed.

Let **a, b, c** be three vectors, given in this order. We shall suppose that they are coinitial and no two are collinear.

Suppose that, **a**, points towards the right of the reader and, **b,** towards the upper part of the paper.

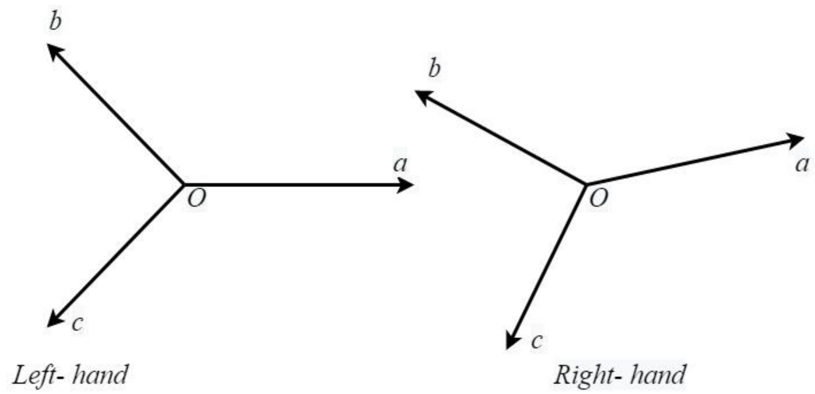


Fig. 2. 1

If now a right-handed or left-handed according as the right-handed screw is translated along **c** or opposite to **c**, when it is rotated from **a** towards **b**, through an angle less than 180^0

It is easily seen that a cyclic permutation of the vectors of a triad does not change the character of the triad from the point of view of its right-handedness and left-handedness. Thus, if **a, b, c** is a right-handed (Left-handed) triad, then

b, c, a and **c, a, b** are also right-handed (left-handed),

but **a, c, b; b, a, c; c, b, a** are left handed (right-handed)

VECTOR PRODUCT OR CROSS PRODUCT

The vector product, denoted by

$$\boldsymbol{a} \times \boldsymbol{b}$$

Of two vectors, **a, b** taken in thus order, is the vector **c,** where

i. $|\boldsymbol{c}| = |\boldsymbol{a}||\boldsymbol{b}| \sin \boldsymbol{\theta}$

θ being the angle between the vectors, **a, b** and $\mathbf{0} \leq \boldsymbol{\theta} \leq \mathbf{180^0}$

ii. *The support of* ***c*** *is perpendicular to the supports of* ***a*** *and* ***b***

iii. *The sense of* ***c*** *is such that the triad* ***a, b, c*** *forms a right-handed system*

As $0 \leq \theta \leq 180^0$, $\sin\theta$ cannot be negative so that $|c|$ will not be negative. Also by definition, the vector product of any vector with the zero vector is the zero vector, *i.e.* ,

$$\boldsymbol{a} \times \mathbf{0} = \mathbf{0}$$

For every vector **a**

It is important to remember that the vector product $\boldsymbol{a} \times \boldsymbol{b}$ is perpendicular to each of the vectors **a** and **b.**

An important Relation

$$(\boldsymbol{a} \times \boldsymbol{b})^{\mathbf{2}} = \boldsymbol{a}^{\mathbf{2}}\boldsymbol{b}^{\mathbf{2}} - (\boldsymbol{a}.\boldsymbol{b})^{\mathbf{2}}$$

We have

$$\begin{aligned}(\boldsymbol{a} \times \boldsymbol{b})^2 &= (\boldsymbol{a} \times \boldsymbol{b}).(\boldsymbol{a} \times \boldsymbol{b})\\ &= (|\boldsymbol{a}||\boldsymbol{b}| \sin \boldsymbol{\theta})^2\\ &= a^2b^2\, sin^2\, \theta\\ &= a^2b^2(1 - cos^2\, \theta)\\ &= a^2b^2 - (|\boldsymbol{a}||\boldsymbol{b}| \cos \theta)^2\\ &= a^2b^2 - (a.b)^2\end{aligned}$$

This relation expresses the length of the vector $\boldsymbol{a} \times \boldsymbol{b}$ in terms of the scalar products

$$\boldsymbol{a}.\boldsymbol{a}, \quad \boldsymbol{b}.\boldsymbol{b}, \quad \boldsymbol{a}.\boldsymbol{b}$$

Formulation of Vector Product in Terms of Scalar Products

The vector product $\boldsymbol{a} \times \boldsymbol{b}$ is the vector, such that

i. $|c| = \sqrt{[a^2b^2] - (a.b)^2}$
ii. $c.a = 0, \quad c.b = 0$
iii. **a, b, c** form a right-handed system

SOME PROPERTIES OF VECTOR PRODUCTS
The Vector Product of two Parallel Vectors is the Zero Vector

In this case, $\theta = 0\ or\ 180^0$, so that $\sin \theta = 0$ and as such $|c| = 0$.

Thus $c = 0$

The Vector Multiplication is not Commutative

In fact

$$\boldsymbol{a} \times \boldsymbol{b} = -\boldsymbol{b} \times \boldsymbol{a}$$

This follows from the fact that while the magnitude and the support of $\boldsymbol{b} \times \boldsymbol{a}$ is the same as those of $\boldsymbol{a} \times \boldsymbol{b}$, their senses are different.

1.1.1. $-\boldsymbol{a} \times \boldsymbol{b} = -(\boldsymbol{a} \times \boldsymbol{b}), \boldsymbol{a} \times (-\boldsymbol{b}) = -(\boldsymbol{a} \times \boldsymbol{b}), (-\boldsymbol{a})(-\boldsymbol{b}) = \boldsymbol{a} \times \boldsymbol{b}$

Generally

$$m\boldsymbol{a} \times n\boldsymbol{b} = mn(\boldsymbol{a} \times \boldsymbol{b}) = \boldsymbol{a} \times mn\boldsymbol{b}$$

Where m and n are any scalars, positive or negative
These results are immediate consequences of the definition of Vector Product

Relations between the mutually perpendicular unit vectors i, j, k

If **i, j, k** are mutually perpendicular unit vectors forming a right-handed system, then, as may be easily seen, from the definition of vector products, we have

$$\boldsymbol{i} \times \boldsymbol{i} = \boldsymbol{0}\ , \boldsymbol{j} \times \boldsymbol{j} = \boldsymbol{0}, \boldsymbol{k} \times \boldsymbol{k} = \boldsymbol{0},$$

$$\boldsymbol{i} \times \boldsymbol{j} = \boldsymbol{k}, \qquad \boldsymbol{j} \times \boldsymbol{i} = -\boldsymbol{k}\ ;$$

$$\boldsymbol{j} \times \boldsymbol{k} = \boldsymbol{i}, \qquad \boldsymbol{j} \times \boldsymbol{i} = -\boldsymbol{k}\ ;$$

$$\boldsymbol{k} \times \boldsymbol{i} = \boldsymbol{j}, \qquad \boldsymbol{i} \times \boldsymbol{k} = -\boldsymbol{j}\ \ ;$$

Note: Later on, we shall show that the Distributive Law also holds for vector multiplications, *i.e.,*
$\boldsymbol{a} \times (\boldsymbol{b} + \boldsymbol{c}) = (\boldsymbol{a} \times \boldsymbol{b}) + (\boldsymbol{a} \times \boldsymbol{c})$ for all vectors **a, b, c**
After establishing the Distributive Law, we shall be able to express

$$(a_1 i + a_2 j + a_3 k) \times (b_1 i + b_2 j + b_3 k)$$

As a linear combination of the vectors**, i, j, k**

Vector Product as a Determinant

Let $a = a_1 i + a_2 j + a_3 k$ and $b = b_1 i + b_2 j + b_3 k$

$$a \times b = a_1 b_1 i \times i + a_1 b_2 i \times j + a_1 b_3 i \times k$$

$$+ a_2 b_1 j \times i + a_2 b_2 j \times j + a_2 b_3 j \times k$$

$$+ a_3 b_1 k \times i + a_3 b_2 k \times j + a_3 b_3 k \times k$$

Then it becomes

$$a \times b = (a_2 b_3 - a_3 b_2) i + (a_3 b_1 - a_1 b_3) j + (a_1 b_2 - a_2 b_1) k$$

This expression can be re-written in the determinant form as

$$a \times b \ = \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

Angle between Two Vectors a and b

We have

$$|\boldsymbol{a} \times \boldsymbol{b}| = |\boldsymbol{a}||\boldsymbol{b}| \sin \boldsymbol{\theta}$$

$$\sin\boldsymbol{\theta} = \frac{|\boldsymbol{a} \times \boldsymbol{b}|}{|\boldsymbol{a}||\boldsymbol{b}|}$$

$$= |\hat{\boldsymbol{a}} \times \hat{b}|$$

In terms of components, it can be easily be shown that

$$\sin\boldsymbol{\theta} = \frac{\sqrt{(a_2\boldsymbol{b_3} - \boldsymbol{a_3b_2})^2 + (\boldsymbol{a_3b_1} - \boldsymbol{a_1b_3})^2 + (\boldsymbol{a_1b_2} - \boldsymbol{a_2b_1})^2}}{\sqrt{(\boldsymbol{a_1}^2 + \boldsymbol{a_2}^2 + \boldsymbol{a_3}^2)} + \sqrt{(\boldsymbol{b_1}^2 + \boldsymbol{b_2}^2 + \boldsymbol{b_3}^2)}}$$

General Form of Vector Product

Let **a** and **b** be any two vectors and l, m, n be three non-coplanar vectors.

Let $\boldsymbol{a} = a_1\boldsymbol{l} + a_2\boldsymbol{m} + a_3\boldsymbol{n}$ and Let $\boldsymbol{b} = b_1\boldsymbol{l} + b_2\boldsymbol{m} + b_3\boldsymbol{n}$ Then

$$\boldsymbol{a} \times \boldsymbol{b} = (a_1\boldsymbol{l} + a_2\boldsymbol{m} + a_3\boldsymbol{n}) \times (b_1\boldsymbol{l} + b_2\boldsymbol{m} + b_3\boldsymbol{n})$$

$$= (a_1b_2 - a_2b_1)\boldsymbol{l} \times \boldsymbol{m} - (a_1b_3 - a_3b_1)\boldsymbol{n} \times \boldsymbol{l} + (a_2b_3 - a_3b_2)\boldsymbol{m} \times \boldsymbol{n}$$

$$= \begin{vmatrix} \boldsymbol{m} \times \boldsymbol{n} & \boldsymbol{n} \times \boldsymbol{l} & \boldsymbol{l} \times \boldsymbol{m} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

EXAMPLES

Determine a unit vector perpendicular to the plane of a and b, where $\boldsymbol{a} = \boldsymbol{4i} + \boldsymbol{3j} - \boldsymbol{k}$ and $\boldsymbol{b} = \boldsymbol{2i} - \boldsymbol{6j} - \boldsymbol{3k}.$ Also obtain sine of the angle between **a** and **b**

Solution

$$\boldsymbol{a} \times \boldsymbol{b} = \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ \boldsymbol{4} & \boldsymbol{3} & \boldsymbol{-1} \\ \boldsymbol{2} & \boldsymbol{-6} & \boldsymbol{-3} \end{vmatrix}$$

$$= \boldsymbol{-15i} + \boldsymbol{10j} - \boldsymbol{30k}$$

As $\boldsymbol{a} \times \boldsymbol{b}$ is the vector perpendicular to the plane of **a** and **b** is

$$= \frac{\boldsymbol{a} \times \boldsymbol{b}}{|\boldsymbol{a} \times \boldsymbol{b}|}$$

$$= \frac{\boldsymbol{-15i} + \boldsymbol{10j} - \boldsymbol{30k}}{\sqrt{(-15)^2 + (10)^2 + (-30)^2}} = -\frac{\boldsymbol{3}}{\boldsymbol{7}}\boldsymbol{i} + \frac{\boldsymbol{2}}{\boldsymbol{7}}\boldsymbol{j} - \frac{\boldsymbol{6}}{\boldsymbol{7}}\boldsymbol{k}$$

If θ be the angle between the vectors, then

$$\sin\theta = \frac{|\boldsymbol{a} \times \boldsymbol{b}|}{|\boldsymbol{a}||\boldsymbol{b}|}$$

$$= \frac{\sqrt{\{(-15)^2 + (10)^2 + (-30)^2\}}}{\sqrt{\{4^2 + 3^2 + (-1)^2\}}\sqrt{\{2^2 + (-6)^2 + (-3)^2\}}}$$

$$= \frac{35}{\sqrt{26}\sqrt{49}}$$

$$= \frac{5\sqrt{26}}{26}$$